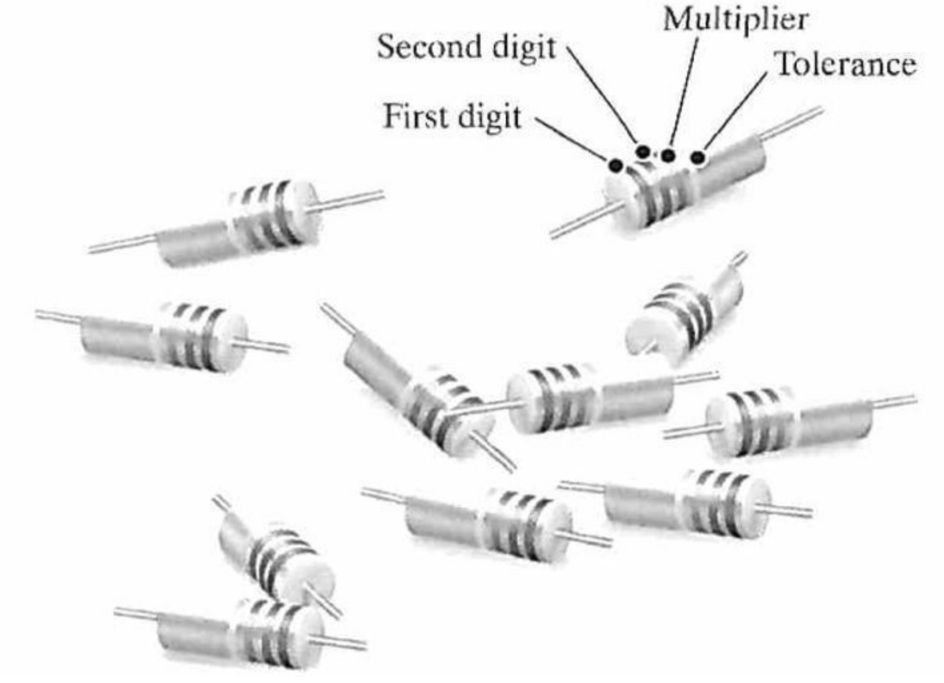
CHAPTER-4

Techniques of Circuit Analysis

Structure

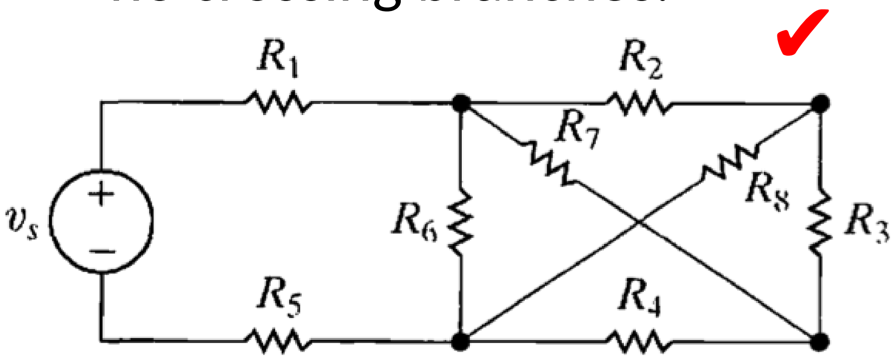
- Terminology
- Node-‐Voltage Method
- Mesh-‐Current Method
- Source Transformation
- Thévenin and Norton Equivalents
- Maximum Power Transfer
- Superpositon

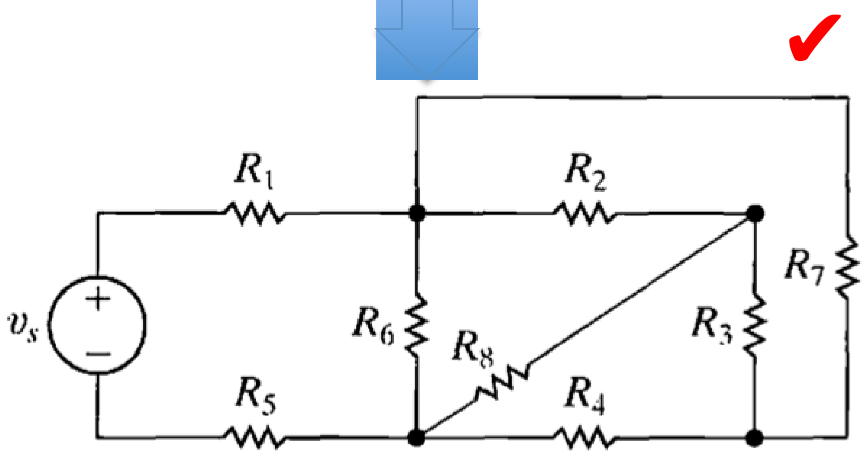
Circuit with Realistic Resistors

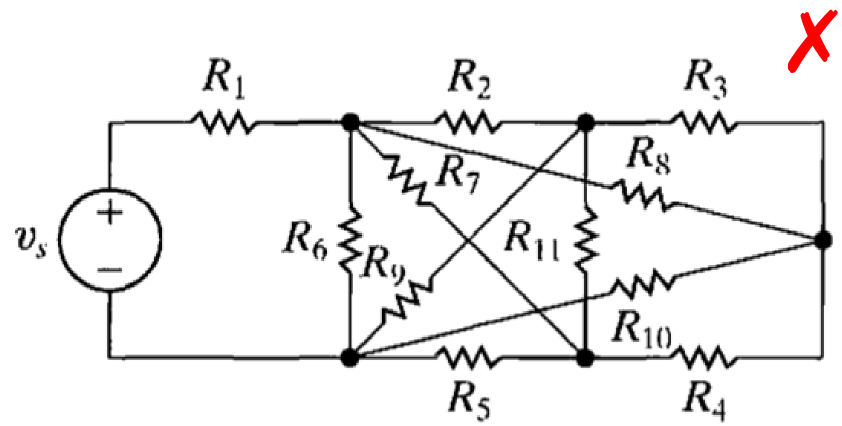


Terminology

Planar circuits: the circuits that can be drawn on a plane with no crossing branches.







More Terms for Describing Circuits

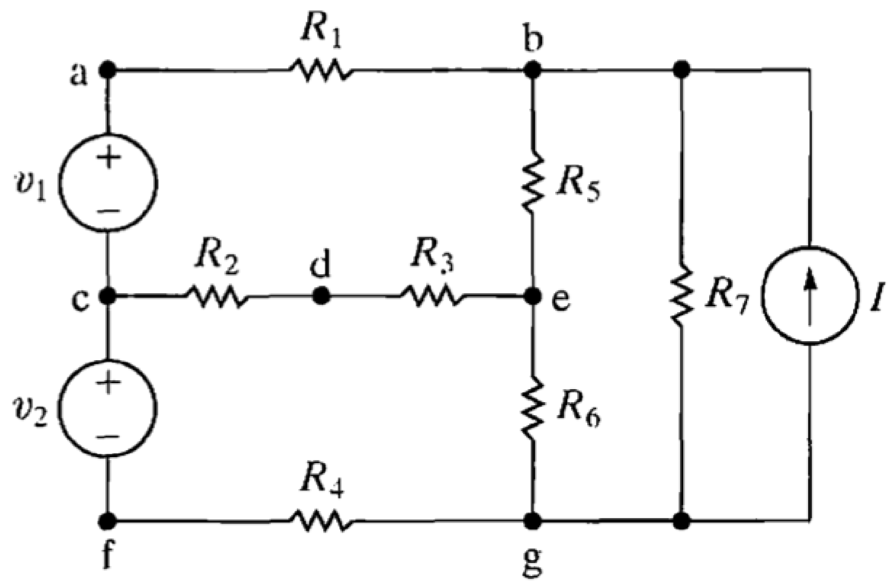
Name	Defini&on
node	A point where two or more circuit elements join
essen:al node	A node where three or more circuit elements join
path	A trace of adjoining basic elements with no elements included more than once
branch	A path that connects two nodes
essen:al branch	A path which connects two essential nodes without passing through an essential node
loop	A path whose last node is the same as the starting node
mesh	A loop that does not enclose any other loops
planar circuit	A circuit that can be drawn on a plane with no crossing branches

Example #1

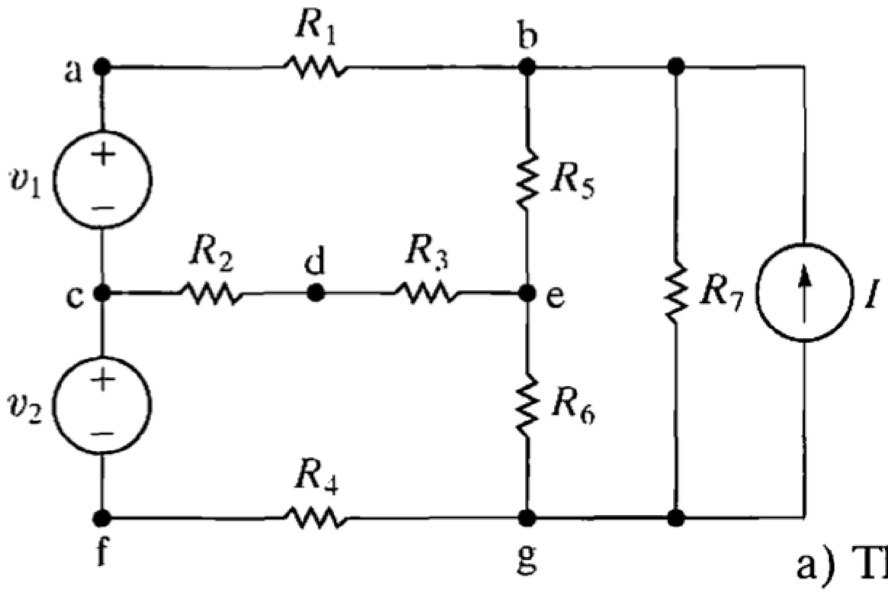
For the circuit in Fig. 4.3, identify:

a) all nodes.
b) all essential nodes.
c) all branches.
d) all essential branches.
e) all meshes.
f) two paths that are not loops or essential branches.

g) two loops that are not meshes.



Soluton for Example #1



a) The nodes are a, b, c, d, e, f, and g.

b) The essential nodes are b, c, e, and g.

c) The branches are v_1, v_2, R_1, R_2, R_3, R_4, R_5, R_6, R_7, and I.

d) The essential branches are $v_1 - R_1$, $R_2 - R_3$, $v_2 - R_4$, R_5, R_6, R_7, and I.

e) The meshes are $v_1 - R_1 - R_5 - R_3 - R_2$, $v_2 - R_2 - R_3 - R_6 - R_4$, $R_5 - R_7 - R_6$, and $R_7 - I$.

Simultaneous Equatons

b: number of branches; n: number of nodes

Kirchhoff's current law: n-1 independent equations

Kirchhoff's voltage law: b-(n-1) independent equations

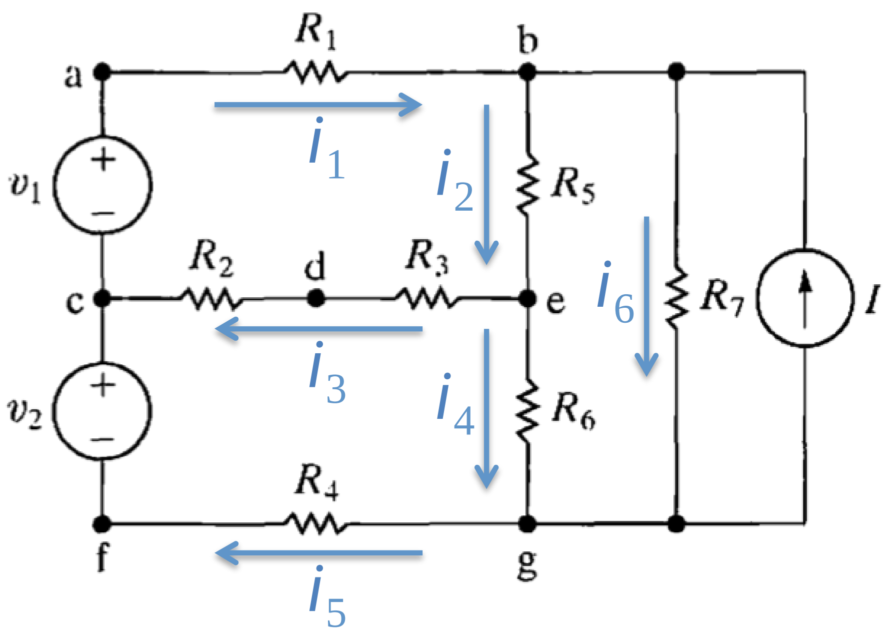
b_e: number of essential branches

n_e: number of essential nodes

Kirchhoff's current law: n_e-1 independent equations

Kirchhoff's voltage law: b_e-(n_e-1) independent equations

Example #2



Seven essential branches

Four essential nodes:

$$-i_1 + i_2 + i_6 - I = 0,$$

$$i_1 - i_3 - i_5 = 0,$$

$$i_3 + i_4 - i_2 = 0.$$

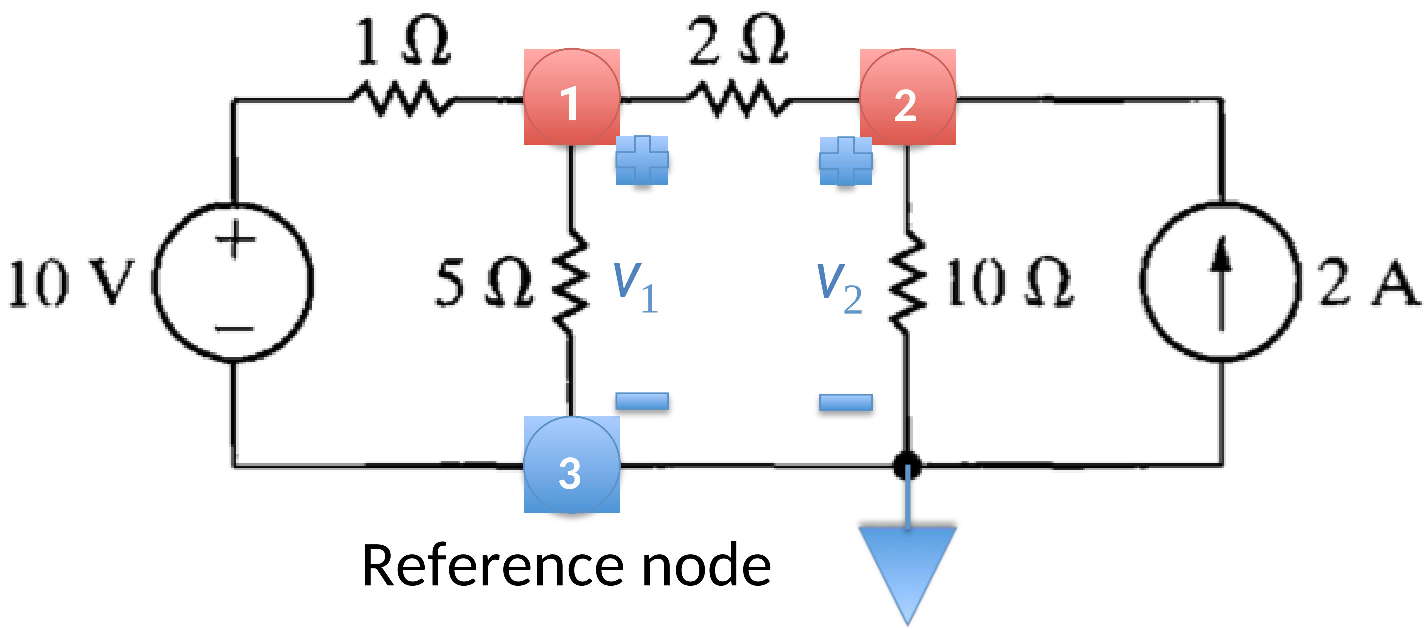
Three meshes:

$$R_1 i_1 + R_5 i_2 + i_3(R_2 + R_3) - v_1 = 0,$$

$$-i_3(R_2 + R_3) + i_4 R_6 + i_5 R_4 - v_2 = 0,$$

$$-i_2 R_5 + i_6 R_7 - i_4 R_6 = 0.$$

Node--Voltage Method

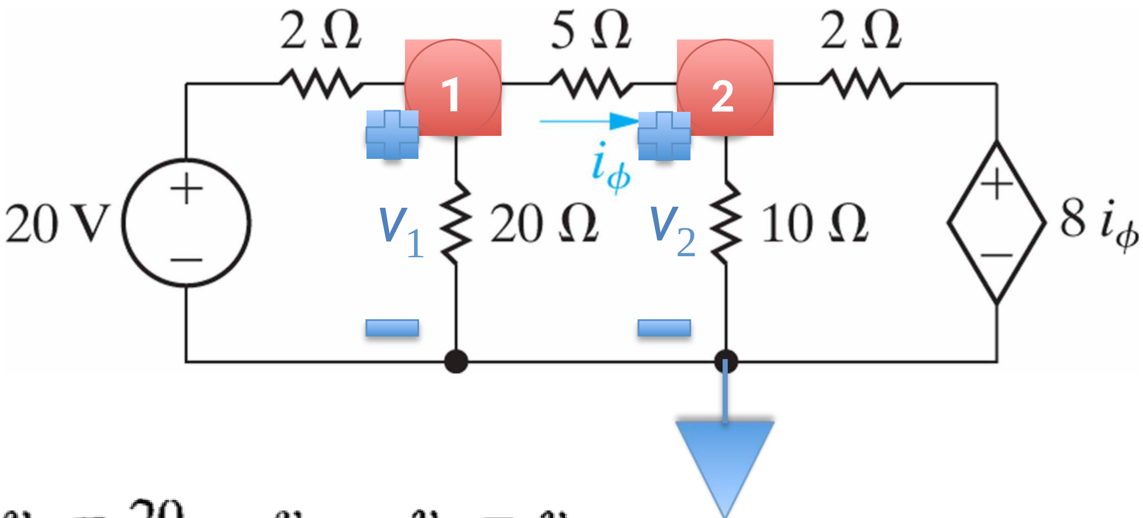


Node 1: $$\frac{v_1 - 10}{1} + \frac{v_1}{5} + \frac{v_1 - v_2}{2} = 0$$

Node 2: $$\frac{v_2 - v_1}{2} + \frac{v_2}{10} - 2 = 0$$

Dependent Sources

Use the node--voltage method to find the power dissipated in the 5Ω resistor in the circuit.

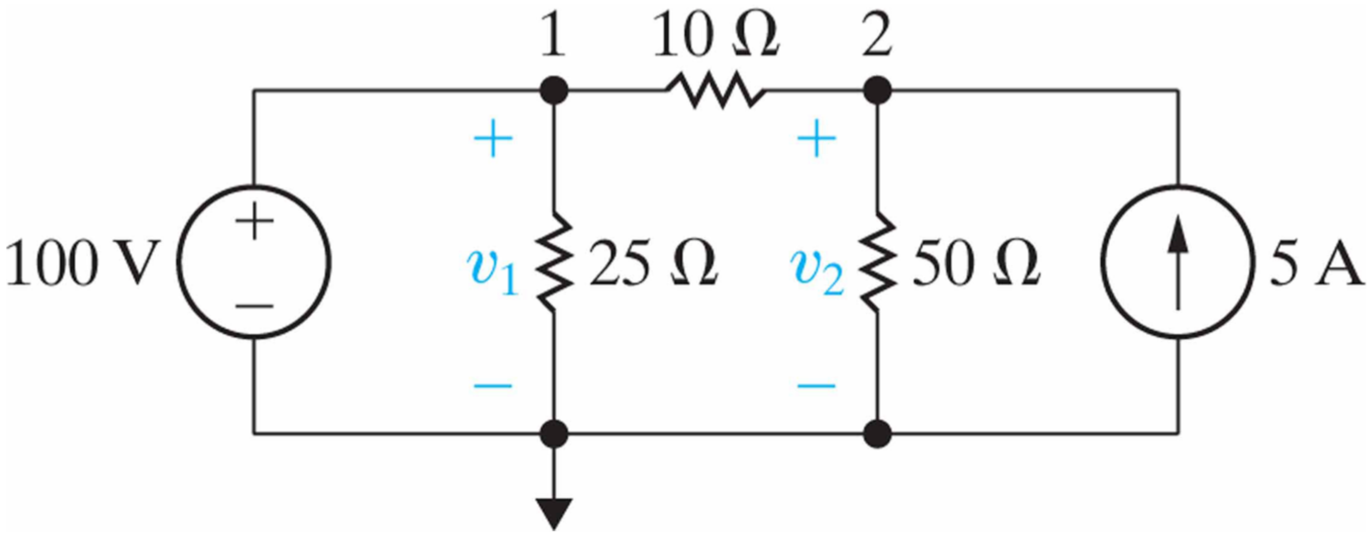


Node 1: $$\frac{v_1 - 20}{2} + \frac{v_1}{20} + \frac{v_1 - v_2}{5} = 0$$

Node 2: $$\frac{v_2 - v_1}{5} + \frac{v_2}{10} + \frac{v_2 - 8i_\phi}{2} = 0$$

$$i_\phi = \frac{v_1 - v_2}{5}$$

Special Case

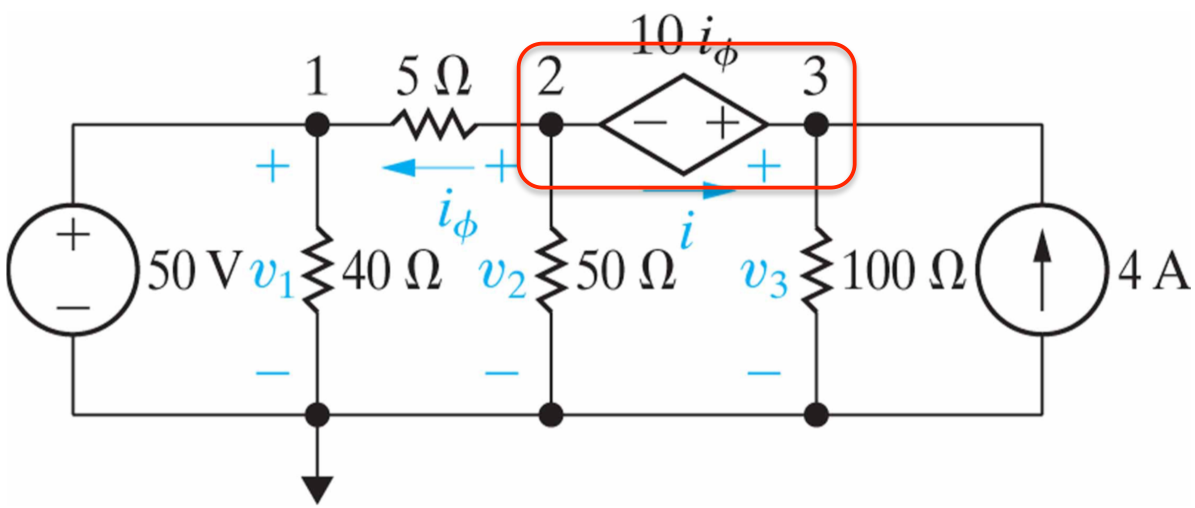


$$\frac{v_2 - v_1}{10} + \frac{v_2}{50} - 5 = 0$$

$$v_1 = 100 \text{ V}$$

$$v_2 = 125 \text{ V}$$

Supernode



Node 2: $\dfrac{v_2 - v_1}{5} + \dfrac{v_2}{50} + i = 0$

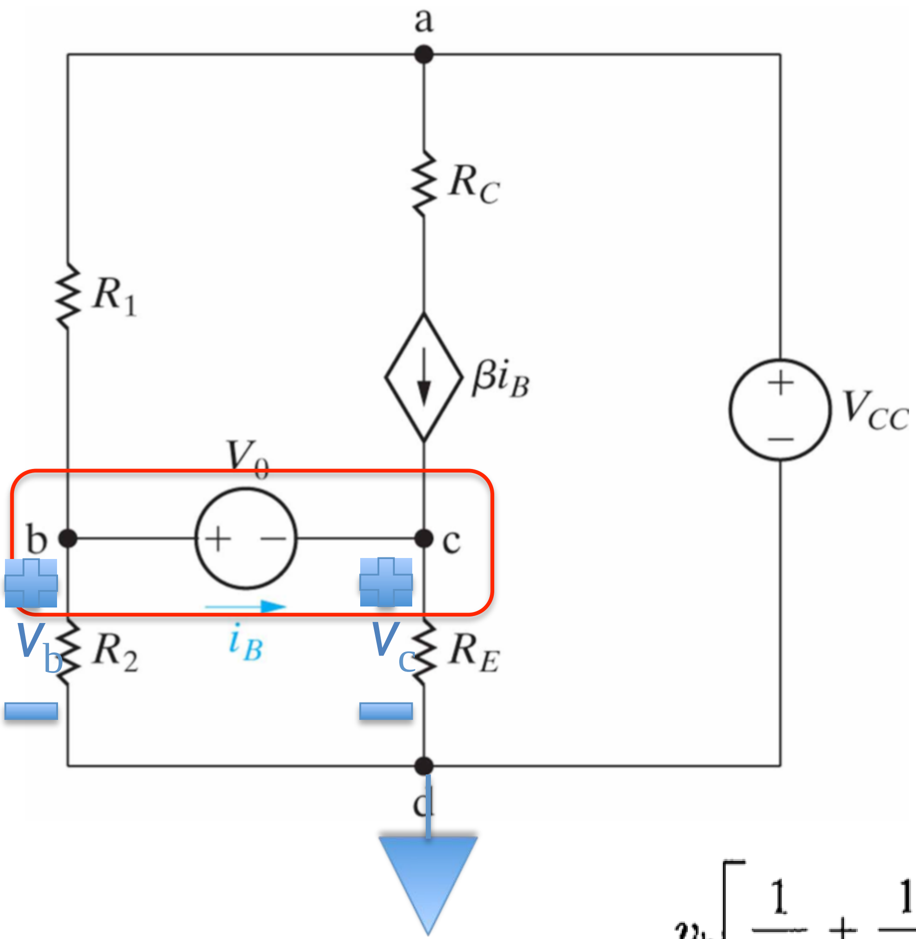
Node 3: $\dfrac{v_3}{100} - i - 4 = 0$

$$\frac{v_2 - v_1}{5} + \frac{v_2}{50} + \frac{v_3}{100} - 4 = 0$$

$$i_\phi = \frac{v_2 - 50}{5}$$

$$v_3 = v_2 + 10i_\phi$$

Example #3



The circuit has four essential nodes: Nodes *a* and *d* are connected by an independent voltage source as are nodes *b* and *c*. Therefore the problem reduces to finding a single unknown node voltage, *because (n_e -1)-2 = 1*

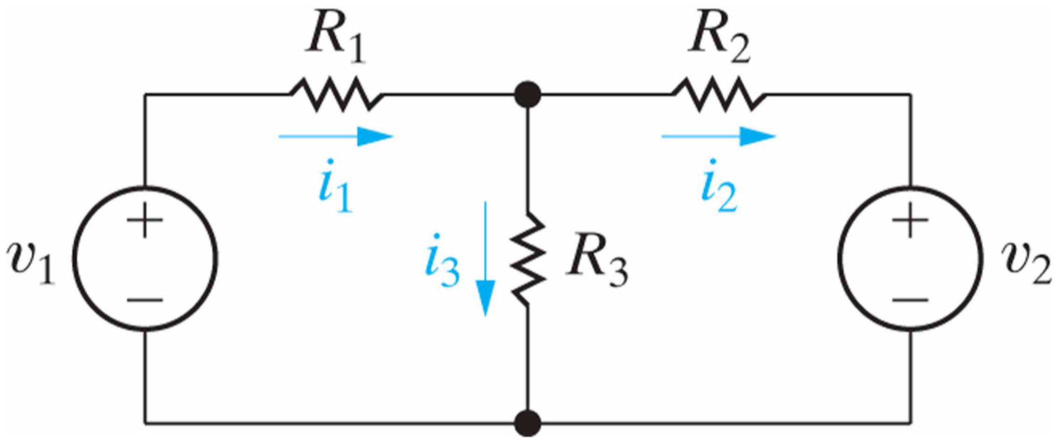
$$\frac{v_b}{R_2} + \frac{v_b - V_{CC}}{R_1} + \frac{v_c}{R_E} - \beta i_B = 0$$

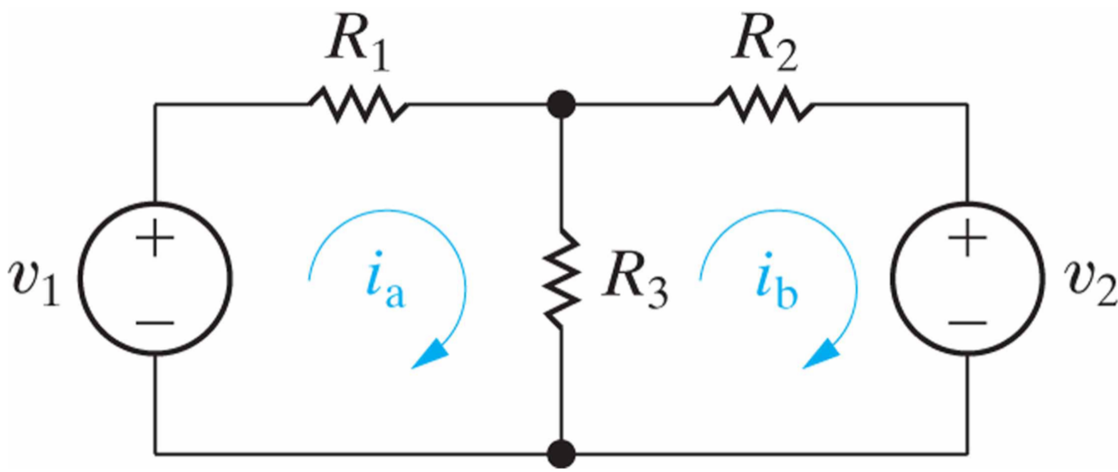
$$v_c = (i_B + \beta i_B)R_E$$

$$v_c = v_b - V_0$$

$$v_b\left[\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{(1+\beta)R_E}\right] = \frac{V_{CC}}{R_1} + \frac{V_0}{(1+\beta)R_E}$$

Mesh-Current Method



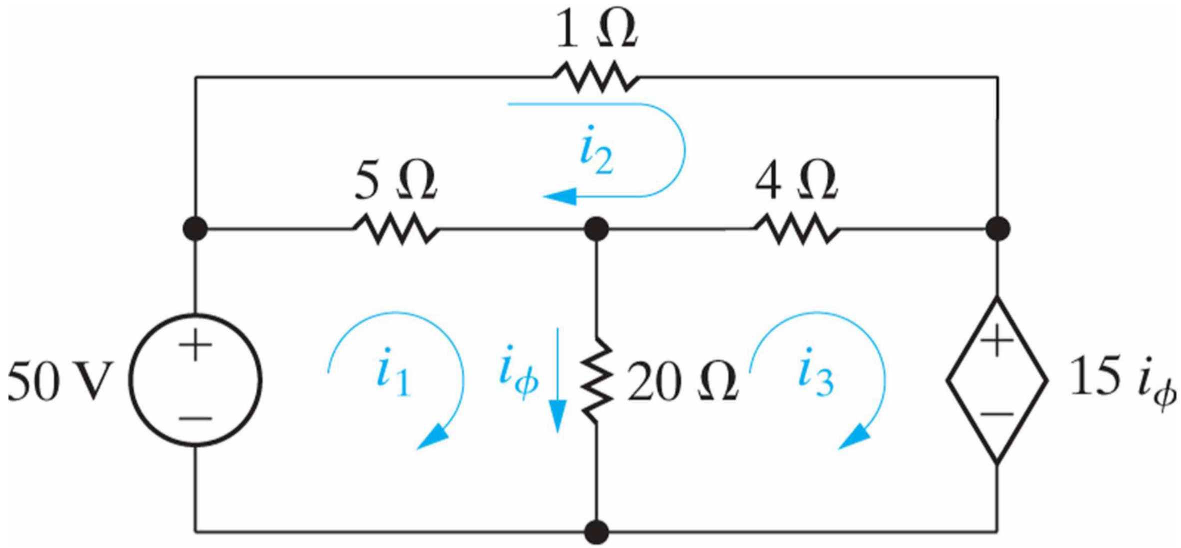


b_e=3 n_e=2 b_e-(n_e-1)=2

$$v_1 = i_a R_1 + (i_a - i_b) R_3$$

$$-v_2 = (i_b - i_a) R_3 + i_b R_2$$

Dependent Sources



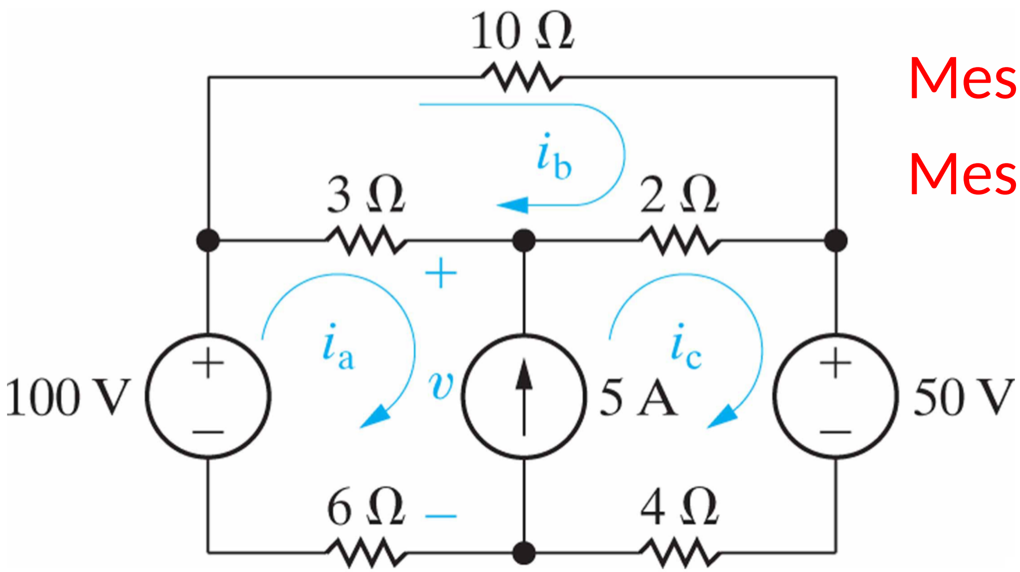
Mesh 1: $50 = 5(i_1 - i_2) + 20(i_1 - i_3)$

Mesh 2: $0 = 5(i_2 - i_1) + 1i_2 + 4(i_2 - i_3)$

Mesh 3: $0 = 20(i_3 - i_1) + 4(i_3 - i_2) + 15i_\phi$

Constraint: $i_\phi = i_1 - i_3$

Special Case



Mesh a: $100 = 3(i_a - i_b) + v + 6i_a$

Mesh c: $-50 = 4i_c - v + 2(i_c - i_b)$

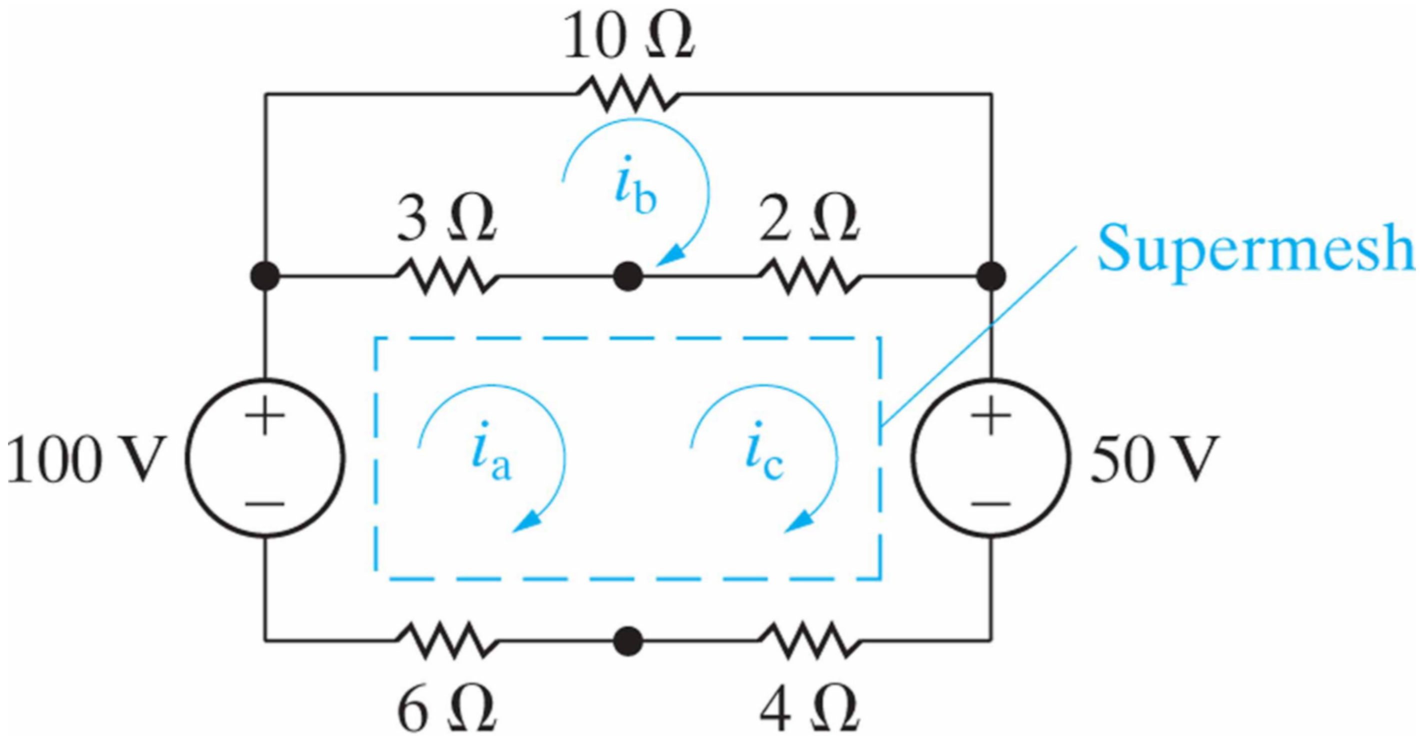
$$50 = 9i_a - 5i_b + 6i_c$$

Mesh b: $0 = 3(i_b - i_a) + 10i_b + 2(i_b - i_c)$

Constraint: $i_c - i_a = 5$

$$i_a = 1.75 \text{ A}, \quad i_b = 1.25 \text{ A}, \quad \text{and} \quad i_c = 6.75 \text{ A}.$$

Supermesh



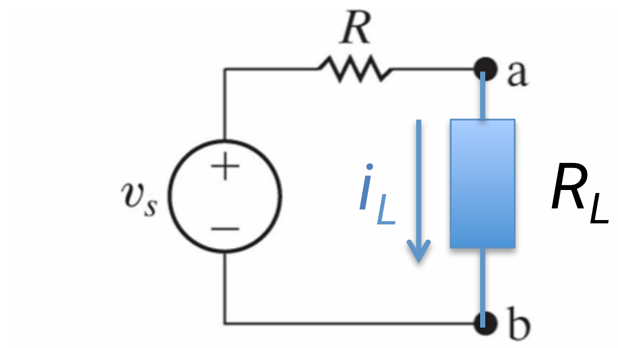
$$-100 + 3(i_a - i_b) + 2(i_c - i_b) + 50 + 4i_c + 6i_a = 0.$$

$$50 = 9i_a - 5i_b + 6i_c.$$

Node--Voltage vs. Mesh--Current

- Does one of the methods result in fewer simultaneous equations to solve?
- Does the circuit contain supernodes? If so, using the node--voltage method will permit you to reduce the number of equations to be solved.
- Does the circuit contain supermeshes? If so, using the mesh--current method will permit you to reduce the number of equations to be solved.
- Will solving some por:on of the circuit give the requested solution? If so, which method is most efficient for solving just the pertinent portion of the circuit?

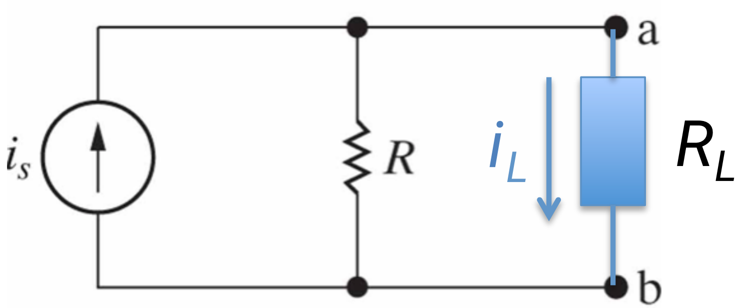
Source Transformatons



(a)

$$i_L = \frac{v_s}{R + R_L}$$

$$i_s = \frac{v_s}{R}$$



(b)

$$i_L = \frac{R}{R + R_L} i_s$$

Equivalent circuits with respect to terminals a and b

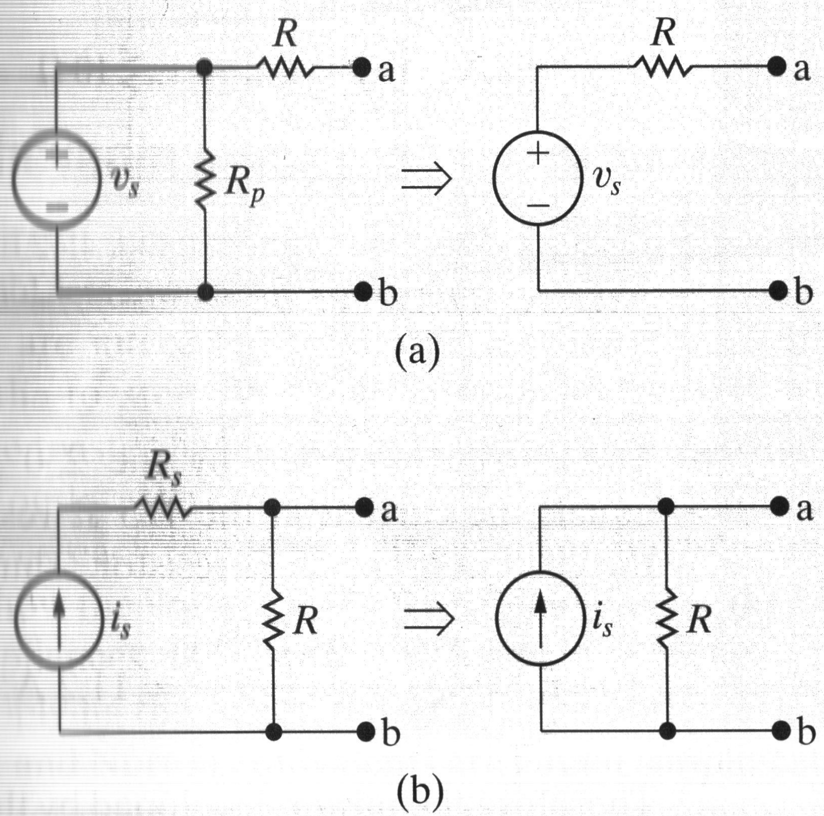


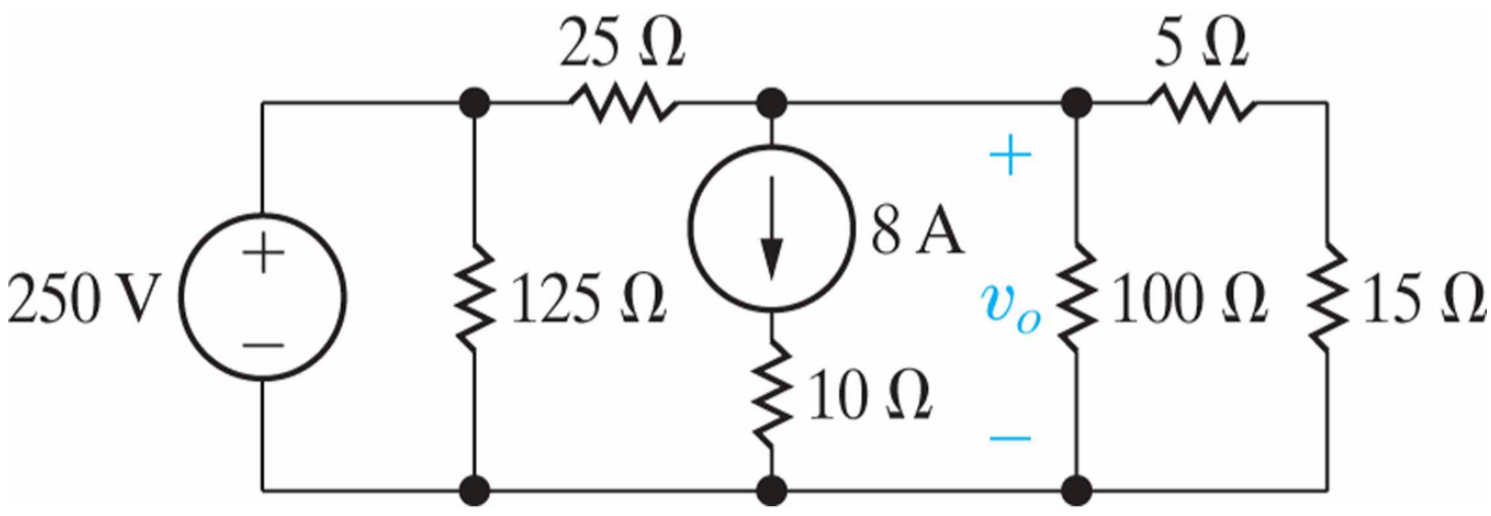
Figure 4.39 Equivalent circuits containing a resistance in parallel with a voltage source or in series with a current source.

Example #4

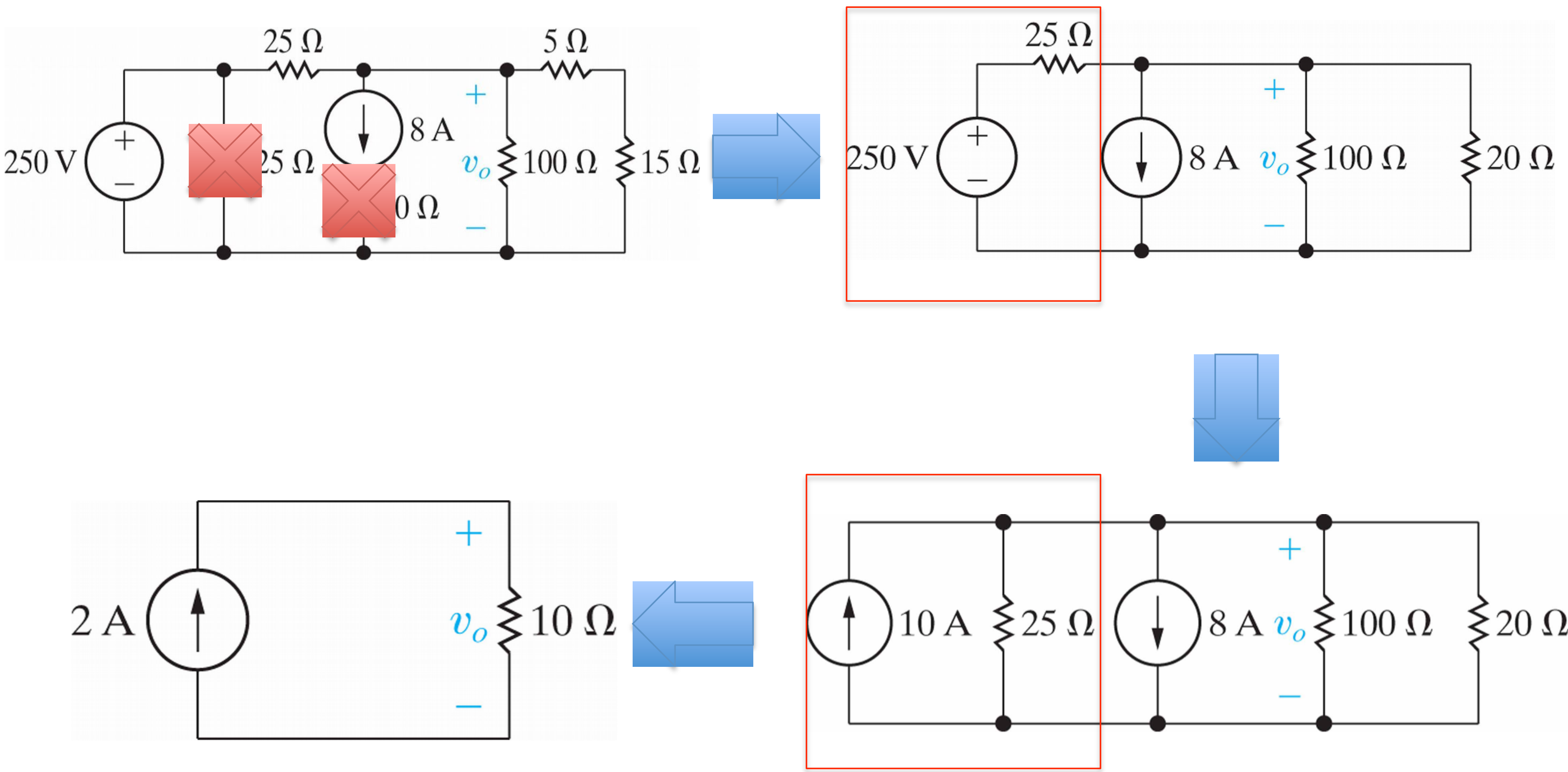
(a) Use source transformations to find the voltage v_o in the circuit.

(b) Find the power developed by the 250 V voltage source.

(c) Find the power developed by the 8 A current source.



Soluton for Example #4

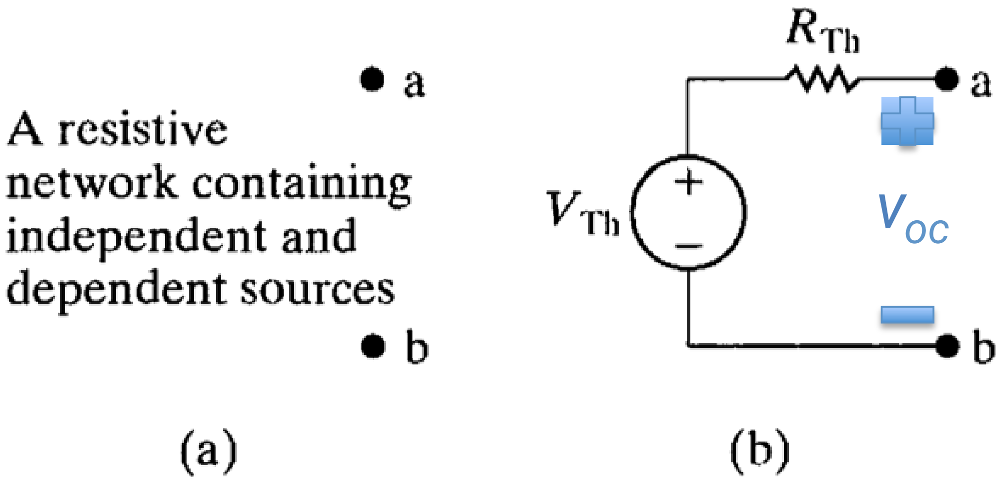


Thévenin and Norton Equivalents

- **Thévenin and Norton equivalents** are circuit simplification techniques that focus on terminal behavior and thus are extremely valuable aids in analysis.

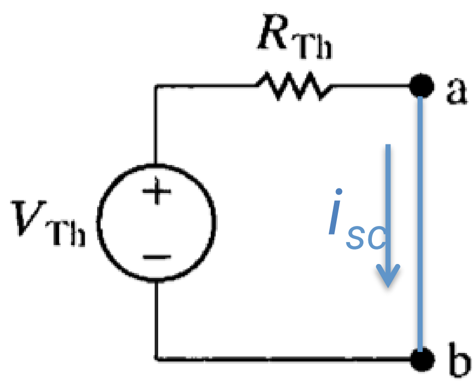
- **Thevenin equivalent circuit:** an independent voltage source V_{Th} in series with a resistor R_{Th}, which replaces an interconnection of sources and resistors.
- **Norton equivalent circuit:** consists of an independent current source in parallel with the Norton equivalent resistance,

Thévenin Equivalent



$V_{Th} = v_{oc}$

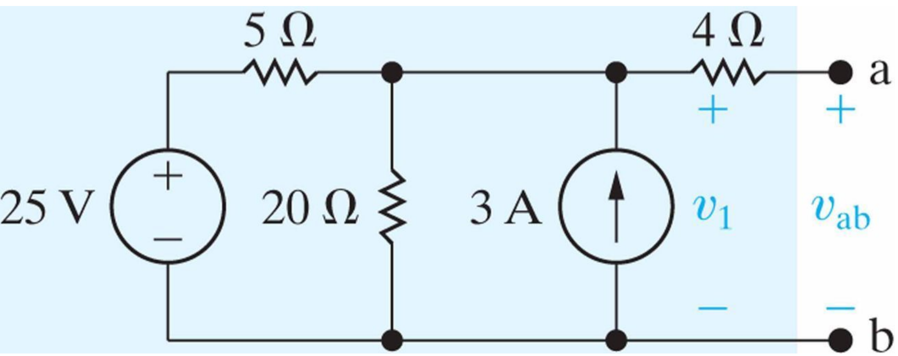
Thévenin voltage V_{Th} equals to the open-circuit voltage in the original circuit.



$R_{Th} = V_{Th}/i_{sc} = v_{oc}/i_{sc}$

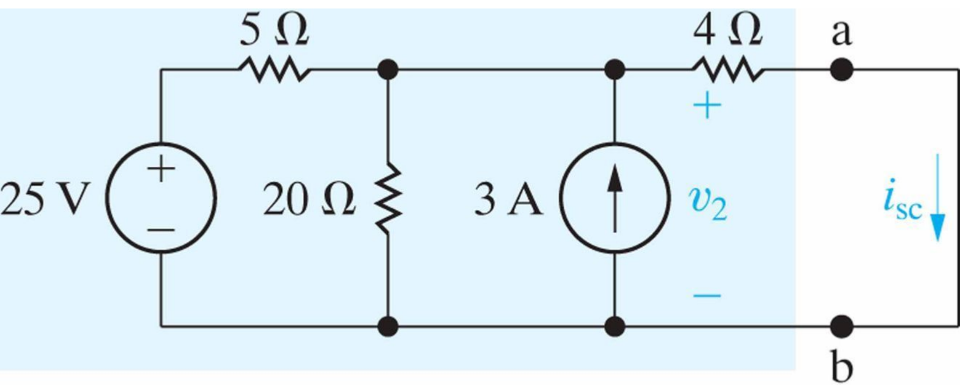
Thévenin resistance R_{Th} is the ratio of the open-circuit voltage to the short-circuit current.

Example #5



$$\frac{v_1 - 25}{5} + \frac{v_1}{20} - 3 = 0$$

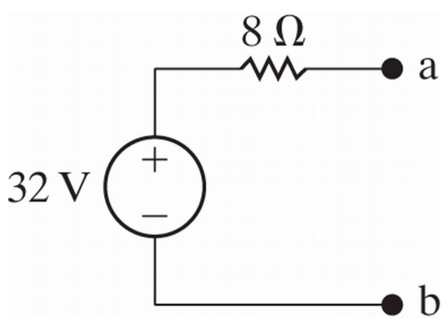
$V_{Th} = v_1 = 32$ V



$$\frac{v_2 - 25}{5} + \frac{v_2}{20} - 3 + \frac{v_2}{4} = 0$$

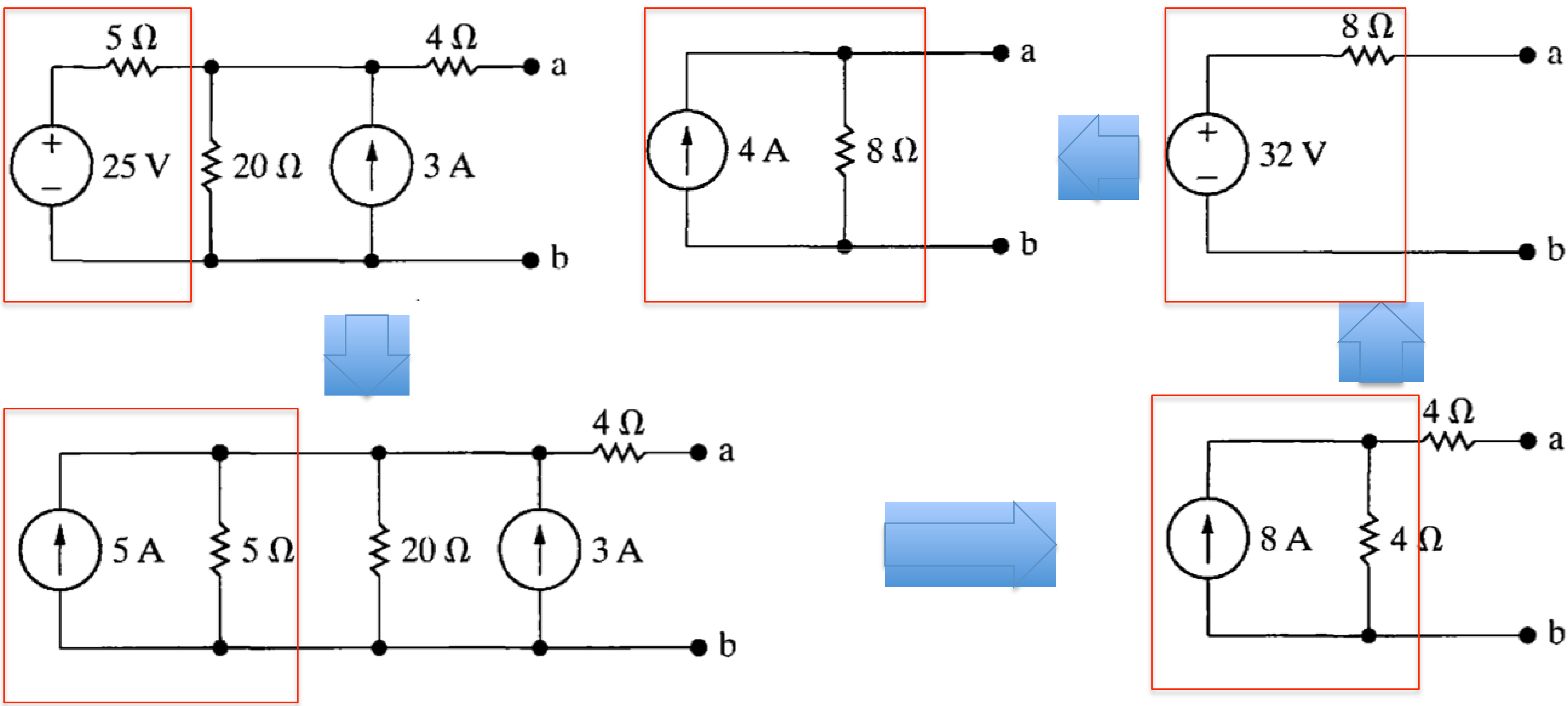
$$v_2 = 16 \text{ V} \qquad i_{sc} = \frac{16}{4} = 4 \text{ A}$$

$R_{Th} = V_{Th}/i_{sc} = 8$ A

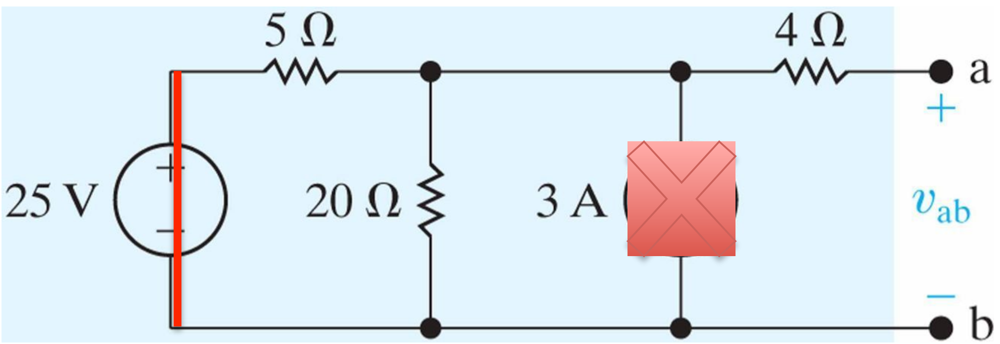


Thévenin Equivalents

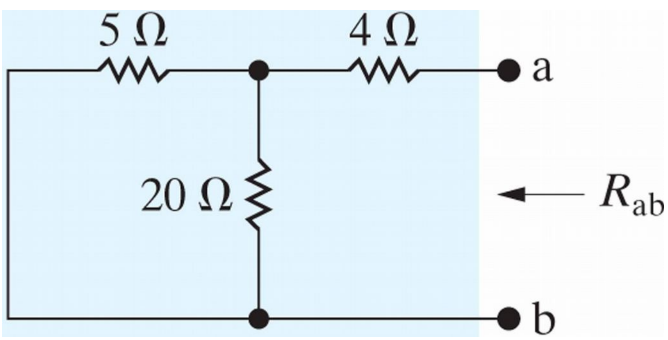
Norton Equivalent



More on Thevenin Equivalent



A voltage source is deactivated by replacing it with a short circuit.
A current source is deactivated by replacing it with an open circuit.

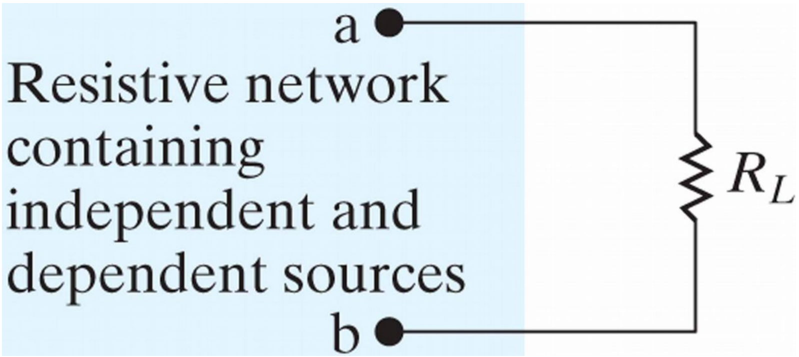


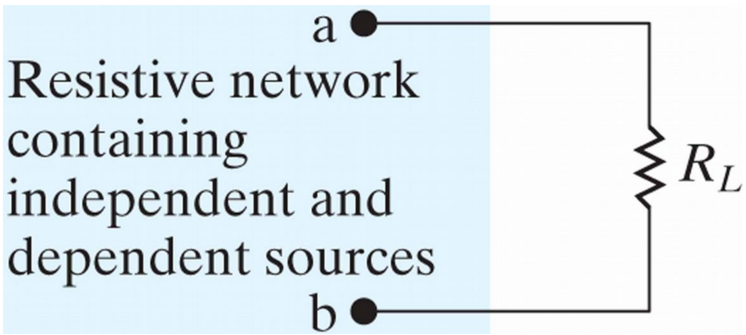
$$R_{ab} = R_{Th} = 4 + \frac{5 \times 20}{25} = 8\ \Omega$$

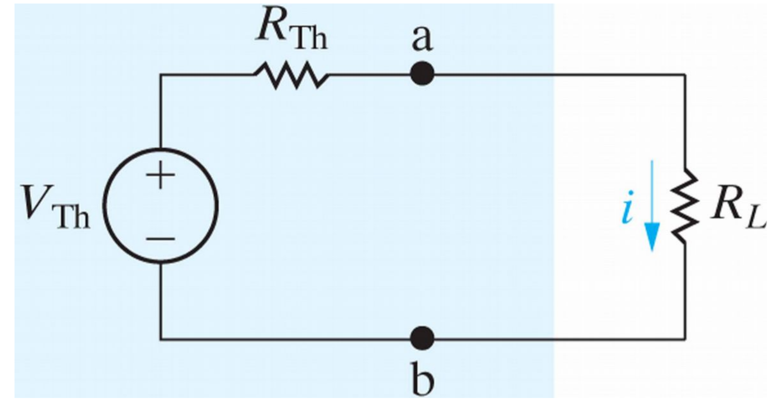
Maximum Power Transfer

Power Transfer

- The first emphasizes the efficiency of the power transfer.
  - Power utility systems are a good example of this type because they are concerned with the generation, transmission, and distribution of large quantities of electric power.
- The second basic type of system emphasizes the amount of power transferred. ✔
  - Communication and instrumentation systems are good examples because in the transmission of information, or data, via electric signals, the power available at the transmitter or detector is limited







$$p = i^2 R_L = \left(\frac{V_{\text{Th}}}{R_{\text{Th}} + R_{\text{L}}} \right)^2 R_L.$$

$$\frac{dp}{dR_L} = V_{\text{Th}}^2 \left[\frac{(R_{\text{Th}} + R_L)^2 - R_L \cdot 2(R_{\text{Th}} + R_L)}{(R_{\text{Th}} + R_L)^4} \right].$$

p is maximized when the derivative is zero, thus we have

$$(R_{\text{Th}} + R_L)^2 = 2R_L(R_{\text{Th}} + R_L)$$

$$R_L = R_{\text{Th}}. \qquad p_{\max} = \frac{V_{\text{Th}}^2 R_L}{(2R_L)^2} = \frac{V_{\text{Th}}^2}{4R_L}$$

Summary

- Basic terms: node, essental node, path, branch, essental branch, mesh, and planar circuit.
- Node-‐voltage and mesh-‐current methods
- Source transformatons
- Thévenin and Norton equivalents
- Maximum power transfer
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